Polybrominated diphenyl ethers in food and human dietary exposure: a review of the recent scientific literature.

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Polybrominated diphenyl ethers (PBDEs) are a class of brominated flame retardants (BFRs) used to protect people from fires by reducing the flammability of combustible materials. In recent years, PBDEs have become widespread environmental pollutants, while body burden in the general population has been increasing. A number of studies have shown that, as for other persistent organic pollutants, dietary intake is one of the main routes of human exposure to PBDEs. The most recent scientific literature concerning the levels of PBDEs in foodstuffs and the human dietary exposure to these BFRs are here reviewed. It has been noted that the available information on human total daily intake through food consumption is basically limited to a number of European countries, USA, China, and Japan. In spite of the considerable methodological differences among studies, the results show notable coincidences such as the important contribution to the sum of total PBDEs of some congeners such as BDEs 47, 49, 99 and 209, the comparatively high contribution of fish and seafood, and dairy products, and the probably limited human health risks derived from dietary exposure to PBDEs. Various issues directly related to human exposure to PBDEs through the diet still need investigation.